

Learning Objective 2.7: I can measure the radiation pattern of an antenna and extract gain, beamwidth, sidelobe level, and polarization from the data.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Is the range adequate?** You are setting up a pattern range at $f = 2.45$ GHz. The antenna under test (AUT) is a pyramidal horn with a 24×17 cm aperture. The lab bench gives you a separation of $r = 3.0$ m.

- (a) Find the wavelength and the AUT's largest dimension D , then evaluate the far-field criterion $r \geq 2D^2/\lambda$.

$$2D^2/\lambda =$$

- (b) The far field also requires $r \geq 5D$ and $r \geq 10\lambda$. Evaluate both, state which criterion binds, and say whether 3.0 m is acceptable.

$$\text{binding } r =$$

- (c) In one or two sentences: if you ran this measurement at $r = 0.5$ m instead, what specifically would go wrong with the recorded pattern, and would the measured HPBW read too wide or too narrow?

- (d) Your calibrated reference horn has a 34×25 cm aperture. Recompute the far-field distance for *it*, and state what that means for the range you just declared acceptable.

$$2D^2/\lambda =$$

2. **Reducing a measured cut.** The table below is one measured E-plane cut of an aperture antenna, recorded as $|S_{21}|$ in dB. The cut was symmetric about 0° to within 0.3 dB, so only $\theta \geq 0$ is tabulated. With the AUT unmated and its cable terminated in $50\ \Omega$, every other setting unchanged, the same sweep read -78.0 dB.

θ	level	θ	level	θ	level
0°	-35.6	18°	-49.5	48°	-57.5
2°	-36.0	20°	-48.7	60°	-59.8
4°	-36.7	22°	-50.0	75°	-62.5
6°	-38.3	24°	-52.7	90°	-64.0
8°	-41.4	28°	-77.6	120°	-66.5
10°	-45.4	32°	-55.6	150°	-63.0
12°	-53.5	36°	-53.3	180°	-60.6
14°	-66.4	40°	-56.7		
16°	-52.6	44°	-70.6		

- (a) Extract the half-power beamwidth. Interpolate between samples – do not snap to the nearest grid point.

$$\theta_{HP} =$$

- (b) Find the first sidelobe level relative to the peak, and compare it with the canonical value for a uniformly illuminated aperture.

$$\text{SLL} =$$

- (c) Compute the front-to-back ratio.

$$\text{F/B} =$$

- (d) The operator used a 2° step inside the main beam. Using the rule of thumb for angle step, was that fine, wasteful, or too coarse? Answer in one sentence, with the number that justifies it.

3. **Gain by comparison, and polarization.** On the range of Question 1 you record a peak of -35.6 dB with the AUT installed. You then swap in the calibrated reference horn ($G_{ref} = 15.0$ dBi), change nothing else, and record -32.4 dB.

(a) Find the gain of the AUT in dBi.

$$G_{AUT} =$$

(b) The procedure insists that you change *nothing* between the two measurements. Name two things that would silently corrupt the result if you changed them, and say what the resulting gain error would look like.

(c) You rotate the source antenna 90° about the range axis and re-peak: the boresight level drops to -59.9 dB. Compute the cross-polarization discrimination.

$$\text{XPD} =$$

(d) Suppose instead the cross-pol peak had landed only 6.0 dB above your noise floor. By how much would the measured cross-pol level overstate the true one, and what does that do to your reported XPD?

$$\text{error} =$$

4. **What the noise floor is doing to your numbers.** A measured pattern reaches its peak at -35.6 dB ; the terminated floor run, taken at the same settings, reads -78.0 dB . Levels below are quoted relative to the peak.

- (a) A null in a different data set measures -25.0 dB on a system whose floor sits at -30.0 dB relative to the peak. Estimate the true depth of that null.

true null =

- (b) Now do the same subtraction for a reading of -42.0 dB on this system, whose floor is at $-78.0 + 35.6 = -42.4$ dB relative. Comment on the answer you get.

true null =

- (c) You average 16 sweeps instead of 1. In one or two sentences, say what improves and what does not, and why.

- (d) State this system's dynamic range, then rank HPBW (12.4°), first sidelobe (-13.1 dB) and null depth (-42.0 dB) by how much you trust them, with the margin above the floor for each.

DR =

5. **The uncertainty paragraph you have to write.** Your report claims $G_{AUT} = 11.8$ dBi for the horn of Question 3, against an aperture-formula prediction of 12.3 dBi (using $\eta_{ap} = 0.5$).

(a) Your peak-find left the AUT 1.5° off boresight on a 12.4° beam. Using a parabolic approximation to the main beam near its peak, how much does the recorded peak read low?

$\Delta =$

(b) A floor reflection arrives 20 dB below the direct path. Find the peak-to-peak ripple it puts on your pattern, then find how far down the reflection would have to be to hold the ripple inside ± 0.2 dB.

ripple =

(c) Combine the error sources – reference tolerance 0.5 dB, ripple 0.9 dB, alignment 0.2 dB, sweep-to-sweep repeatability 0.3 dB – into one uncertainty, then write the two-sentence verdict on the 0.5 dB gap between your measurement and the prediction.

Documentation: _____